

Project Design Phase

Proposed Solution Template

Date	31 July 2026
Name	Kesireddy Padma
Project Name	Metro Ticket Generating System in ServiceNow
Maximum Marks	2 Marks

1. Problem Statement (Problem to be solved)

Traditional metro ticket booking can involve long queues, manual ticket handling, difficulty in checking fare details, and delays during busy hours. Passengers need a faster and more convenient way to book tickets and receive their travel information. The Metro Ticket Generating System in ServiceNow addresses these problems by providing a centralized digital ticket-booking solution. It allows passengers to select source and destination stations, choose the ticket type and number of tickets, calculate the fare, complete payment, and receive a digital QR-based ticket. This reduces manual work, minimizes booking errors, and improves the overall passenger experience.

2. Idea / Solution Description

A centralized metro ticket booking application is developed on the ServiceNow platform using a Service Catalog item. The catalog form collects important details such as source station, destination station, passenger type, number of tickets, ticket type, and payment information. Client Scripts and form logic can dynamically validate inputs and calculate the applicable fare. Station details are maintained in a dedicated Metro Station Details table, allowing station information to be mapped to the booking. After successful booking and payment, the system generates a digital QR code and records booking date and ticket expiry information. The solution provides a simple and accessible way for passengers to generate metro tickets digitally.

3. Novelty / Uniqueness

The uniqueness of the solution is the integration of metro ticket generation with ServiceNow's service-management capabilities. Instead of treating ticket booking as a simple static form, the system uses dynamic catalog variables, client-side validation, automated fare calculation, station mapping, payment status tracking, and QR-based digital ticket generation. The combination of these features creates an automated end-to-end ticket-booking process and reduces the need for manual ticket issuance.

4. Social Impact / Customer Satisfaction

The system improves passenger convenience by reducing waiting time at ticket counters and making ticket booking easier. Digital QR tickets provide a quick way to access ticket information and support a more contactless travel experience. Clear booking details, automated fare calculation, and ticket expiry information can reduce confusion and errors. The solution can therefore improve customer satisfaction, especially during peak travel periods, while encouraging digital adoption in public transportation.

5. Business Model (Revenue / Value Model)

The primary value of the solution is operational cost reduction and improved service efficiency. Automating ticket booking, fare calculation, payment-status tracking, and QRticket generation reduces manual effort for metro staff and minimizes errors in ticket processing. The system can also support future integration with digital payment methods and smart-card recharge services. For a metro operator, faster processing and reduced manual intervention can improve service capacity and operational efficiency.

6. Scalability of the Solution

The solution is designed on the ServiceNow platform, allowing it to be extended as the metro network grows. Additional stations, metro lines, cities, passenger categories, and ticket types can be added to the station-details data without redesigning the complete application. The system can also be enhanced with additional payment options, smartcard recharge, notifications, reporting, analytics, and mobile-friendly access. This makes the solution suitable for expansion from a small station network to a larger metroticketing environment.